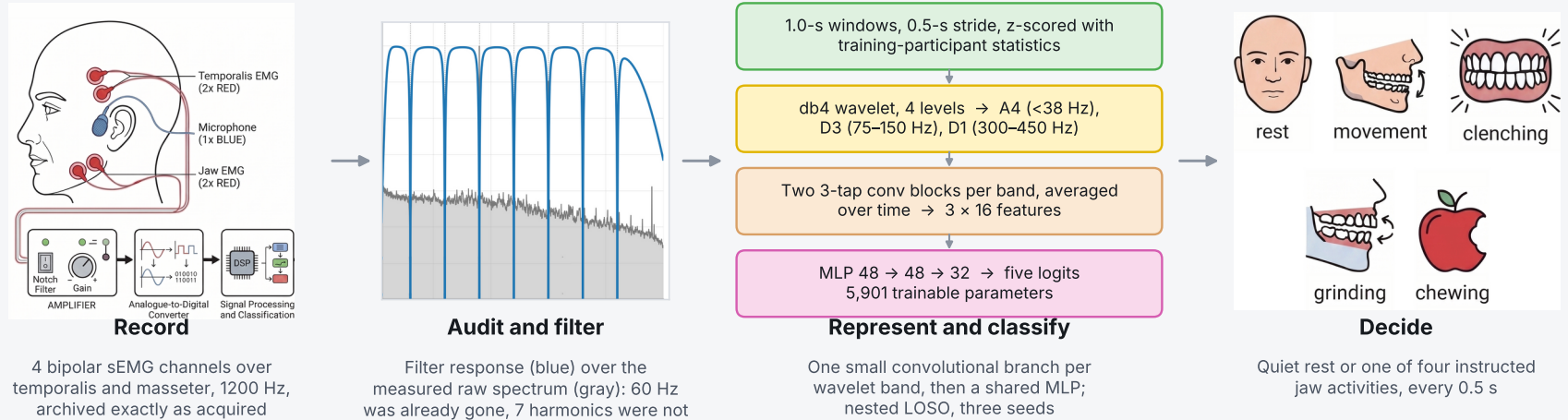


Signal Quality, Representation, and Pre-Modelling Choices

A measured performance budget for surface-EMG jaw-activity classification on unseen participants

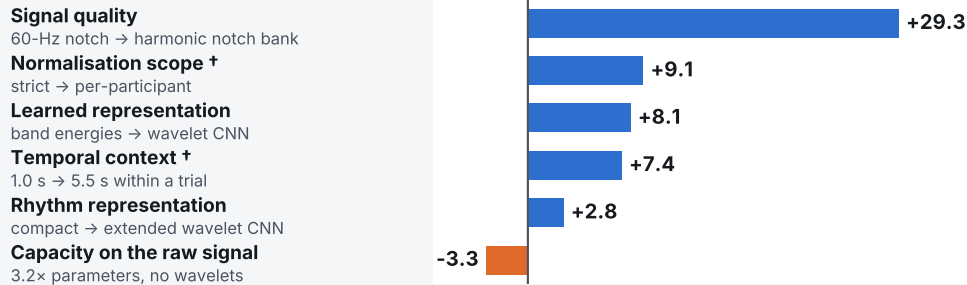
5 adults with diagnosed bruxism · 100 one-minute recordings · 6,173 one-second windows · quiet rest and 4 instructed jaw activities · nested leave-one-subject-out validation

THE PIPELINE, FROM SKIN TO DECISION



WHAT EACH CHOICE WAS WORTH

Change in participant-level macro-F1 (points); each stage varied on identical windows, folds and labels



[†] Legitimate but used by no reported number: presumes a per-user fitting session, or aggregates within a single-condition trial. Rows are not additive; three are screening measurements, reliable in direction and optimistic in level.

ON PARTICIPANTS THE MODEL NEVER SAW

81.7% accuracy
72.0% macro-F1
0.958 macro AUC

means over five held-out participants and three seeds; 5,901 parameters

	true					
	Rest	Movement	Clenching	Grinding	Chewing	
Rest	0.91	0.02	0.00	0.07	0.00	
Movement	0.10	0.73	0.14	0.00	0.03	
Clenching	0.01	0.10	0.68	0.20	0.01	
Grinding	0.07	0.01	0.16	0.72	0.04	
Chewing	0.00	0.10	0.01	0.01	0.88	
	Rest	Movement	Clenching	Grinding	Chewing	predicted

Recall per class, one model per fold.

Quiet rest is recognised in 91% of its windows. Clenching and grinding are the hard pair: 20% and 16% cross over. Movement, the smallest class, absorbs chewing windows, so its F1 (46%) is a precision problem, not a recall one.

Signal quality set the ceiling

The best model on the uncorrected signal (43.5% macro-F1) scored below the weakest baseline on the corrected one (61.3%). Correcting the filter chain was worth 29.3 points.

Representation mattered more than size

Both wavelet models placed above both raw-signal models; $3.2\times$ the parameters on the raw signal lost 3.3 points. Temporal context and normalisation scope were worth as much as the architecture or more.

What transfers is the procedure

Audit the signal against what the analysis assumes, publish the interference statistic beside the accuracy, and price the choices made before modelling against the model.